



Chemical kinetic study of a novel lignocellulosic biofuel: Di-n-butyl ether oxidation in a laminar flow reactor and flames



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ABSTRACT

The combustion characteristics of promising alternative fuels have been studied extensively in the recent years. Nevertheless, the pyrolysis and oxidation kinetics for many oxygenated fuels are not well characterized compared to those of hydrocarbons. In the present investigation, the first chemical kinetic study of a long-chain linear symmetric ether, di-n-butyl ether (DBE), is presented and a detailed reaction model is developed. DBE has been identified recently as a candidate biofuel produced from lignocellulosic biomass. The model includes both high temperature and low temperature reaction pathways with reaction rates generated using appropriate rate rules. In addition, experimental studies on fundamental combustion characteristics, such as ignition delay times and laminar flame speeds have been performed. A laminar flow reactor was used to determine the ignition delay times of lean and stoichiometric DBE/air mixtures. The laminar flame speeds of DBE/air mixtures were measured in the stagnation flame configuration for a wide range of equivalence ratios at atmospheric pressure and an unburned reactant temperature of 373 K. All experimental data were modeled using the present kinetic model. The agreement between measured and computed results is satisfactory, and the model was used to elucidate the oxidation pathways of DBE. The dissociation of keto-hydroperoxides, leading to radical chain branching was found to dominate the ignition of DBE in the low temperature regime. The results of the present numerical and experimental study of the oxidation of di-n-butyl ether provide a good basis for further investigation of long chain linear and branched ethers.

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1. Introduction

Limited fossil fuel reserves, increasing petroleum fuel prices, and the adverse environmental impacts of combustion generated pollutants have motivated research in the recent years related to the utilization of alternative fuels in practical combustion devices. In the transportation sector [1–4], ethers are widely regarded as alternative fuels and/or fuel blending components for compression-ignition engines. Compared with diesel fuels, their high cetane number (CN) and oxygenated molecular structure could result in improved fuel efficiency and reduced pollutant emissions from internal combustion engines. Furthermore, the potential of ether production from lignocellulosic biomass is a significant advantage [5], as that route has no direct competition with food

production. Therefore, ethers are generally considered as important next-generation biofuels [6].

Industrially, ethers can be obtained by dehydration of alcohols [5]. From lignocellulosic biomass, ethanol is produced by hexose fermentation. While ethanol can be converted directly into diethyl ether (DEE) by dehydration, it can dehydrogenate also into acetaldehyde, followed by aldol condensation into crotonaldehyde and subsequent hydrogenation into *n*-butanol. Through the further dehydration of *n*-butanol, di-n-butyl ether (DBE) can be obtained [5]. An alternative method for production of DBE is demonstrated by using dry *n*-butanol derived from fermentation broth [7].

Up to now, combustion research has focused mainly on short chain ethers. For the thoroughly studied dimethyl ether (DME), fundamental investigations on combustion characteristics [8–15], kinetic modeling [13,16,17] as well as practical applications in combustion engines [18,19] can be found in the literature. Recently, Yasunaga et al. [20] studied the pyrolysis and oxidation of diethyl ether (DEE) behind reflected shock waves, presented ignition and species data, and constructed a chemical kinetic model

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to predict the data. The model was later advanced to include the combustion chemistry of branched ethers, ethyl methyl ether (EME), methyl tert-butyl ether (MTBE) and ethyl tert-butyl ether (ETBE) [21]. From the standpoint of quantum chemistry, Di Tommaso et al. [22] conducted Density Functional Theory (DFT) calculations of auto-oxidation reactions of DEE and revealed that the typical competing chain branching and termination pathways found in *n*-alkane oxidation are also dominant in DEE oxidation.

DBE has been identified recently as a candidate biofuel produced from lignocellulosic biomass [5]. Its particularly high cetane number of 100 [3] indicates shorter ignition delay times than those of diesel fuels, which results in a relatively longer combustion process and thus less unburned fuel in compression-ignition engines. Its boiling temperature of 142.4 °C permits facile vaporization of the fuel after injection while minimizing tank evaporation losses. More importantly, high molecular weight fuels are generally attractive as alternative fuels, because their volumetric energy content is comparable to that of petroleum fuels (~36 MJ/L) and they can be applied in conventional compression-ignition engines without the need for extensive engine modifications. While the volumetric energy content of DME is about half that of petroleum fuels [23], the calculated volumetric energy density of DBE and DEE are 31.6 and 26.9 MJ/L, respectively. However, limited combustion studies are reported in the literature for DBE. Beatrice et al. [3] tested DBE as pure fuel in diesel engines and observed a strong decrease of soot emission in exhausts, as expected. Heuser et al. [4] studied DBE as an ignition enhancing blending component for 2-methyltetrahydrofuran (2-MTHF) in single-cylinder diesel engine experiments and found that the addition of DBE to 2-MTHF increases the fuel ignitability, while decreasing the combustion noise level. In addition, CO, unburned hydrocarbons (HC), and soot emissions are reduced notably in comparison to conventional diesel fuel combustion.

The rate parameters for the reactions of OH radical with DBE were investigated by Mellouki et al. [24] at temperatures and pressures relevant to atmospheric oxidation chemistry. In addition, Marrouni et al. [25] carried out density functional theory (DFT) calculations for linear symmetrical monoethers spanning from DME to DBE. The authors reported that rate constants of H-atom abstraction by molecular O₂ are similar for different types of C–H sites. On the other hand, no attempt has been made to determine experimentally the fundamental combustion properties of DBE, e.g. ignition delay times. This has made it challenging to develop the theoretical reaction mechanism, as no experimental data from the literature were available for model validation. At present, computational fluid dynamics (CFD) simulations coupled with detailed chemistry are often performed in research and in engine design to provide insights into fuel combustion in complex combustion devices [26–28]. Despite the combined uncertainties of CFD modeling and chemistry, detailed kinetic models provide higher chemical fidelity in numerical studies compared to models represented by global reactions and empirical equations. However, for DBE, this is presently impossible, since a kinetic model is currently unavailable.

The main goal of this study was to investigate systematically the combustion characteristics of DBE, for the first time. Fundamental combustion properties, such as ignition delay times and laminar flame speeds were determined experimentally. A detailed kinetic model coupled with thermodynamic and transport data was developed based on reaction classes and rate constant rules, and the model was validated against the present data. Both low- and high-temperature kinetic schemes were included in the model. Particular emphasis was placed on low-temperature ignition, which is of great interest to compression-ignition and low temperature combustion (LTC) engines. The kinetic model proposed here was employed to elucidate the oxidation pathways of DBE at rele-

vant conditions. Comparisons of its combustion characteristics with those of *n*-alkanes, alcohols, and short chain ethers were also performed to illustrate the effect of the ether functional group.

The presentation is organized as follows. First, the development of the kinetic model for DBE oxidation is presented. Next, the experimental facilities are described. Then, numerical simulations using the proposed model are presented and compared with the data. Finally, the reaction pathways of DBE oxidation at relevant conditions are illustrated and sensitivity analyses of elementary reactions are performed to provide insights into DBE oxidation chemistry.

2. Kinetic model development

DBE contains an oxygen molecule bound within two normal butyl chains, thus bringing about different associated rate constants, product channels, and higher ignition propensity compared to alkanes, alcohols, and short di-alkyl ethers. This section describes the formulation of a chemical kinetic model for DBE using current understanding of combustion chemistry for a wide variety of structurally similar molecules, such as alcohols, esters, other di-alkyl ethers (e.g., dimethyl ether, diethyl ether, etc.), and *n*-alkanes.

2.1. Naming of species

To illustrate the naming of the species for the model, DBE is denoted as C₄H₉OC₄H₉ in the model (see Fig. 1 for its molecular structure). The carbon chain is labeled numerically (i.e., 1, 2, 3) starting with the location of the carbon bonded to the oxygen. In DBE, the location of a double bond is identified by a hyphen followed by the number of the second carbon in the double bond. Additional notations are provided to denote radical sites in the molecule, where the carbon sites are labeled alphabetically (i.e., a, b, c) starting with the location of the carbon bonded to the oxygen. In this way, the DBE radical at the alpha site (i.e., carbon bonded to the oxygen site) is denoted by C₄H₉OC₄H₈-a.

2.2. General model features

The proposed DBE kinetic model includes both low-temperature and high-temperature kinetics. The DBE model was added to the *n*-butanol model developed recently by Sarathy et al. [29]. The thermodynamic data for DBE and related radicals were calculated using the THERM program of Ritter and Bozzelli [30]. Transport properties for molecules in the high temperature mechanism were determined using the methods described previously in Refs. [31,32], which was developed originally by Tee et al. [33] and Wang and Frenklach [34]. The entire model consists of 426 species and 2335 reactions. 301 species among 1830 reactions are found in the high temperature part. These input files for numerical simulation are available as supplementary material.

The present kinetic model is based on the hierarchical nature of combustion mechanisms (e.g. [32,35,36]). The validation simulations were conducted in CHEMKIN PRO [37] or in FlameMaster [38] using the appropriate reactor modules. A few rare instances required minor modifications to the solver convergence criteria (i.e., absolute tolerance), and in such cases it was ensured that the solution was well converged and accurate.

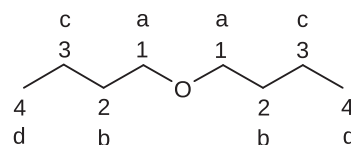


Fig. 1. Structure of di-*n*-butyl ether (C₄H₉OC₄H₉).

The proposed model is presented by going through the various reaction classes and describing the selection of the corresponding rate constants. The classes of elementary reactions considered for the oxidation of DBE are as follows:

High-temperature:

1. Unimolecular fuel decomposition
 2. H-atom abstraction from the fuel
 3. Fuel radical decomposition
 4. Fuel radical isomerization
 5. H-atom abstraction reactions from unsaturated ethers
 6. Addition of radical species O and OH to unsaturated ethers
 7. Reactions of unsaturated ether radicals with HO_2 , CH_3O_2 , and $\text{C}_2\text{H}_5\text{O}_2$
 8. Unsaturated ether radical decomposition
 9. Unimolecular decomposition of unsaturated ethers
- Low-temperature (R refers to a DBE radical):
10. Addition of O_2 to fuel radicals ($\text{R} + \text{O}_2 = \text{ROO}$)
 11. $\text{R} + \text{ROO} = \text{RO} + \text{RO}$
 12. $\text{R} + \text{HO}_2 = \text{RO} + \text{OH}$
 13. $\text{R} + \text{CH}_3\text{O}_2 = \text{RO} + \text{CH}_3\text{O}$
 14. ROO radical isomerization ($\text{ROO} = \text{QOOH}$)
 15. Concerted eliminations ($\text{ROO} = \text{unsaturated ether} + \text{HO}_2$)
 16. $\text{ROO} + \text{HO}_2 = \text{ROOH} + \text{O}_2$
 17. $\text{ROO} + \text{H}_2\text{O}_2 = \text{ROOH} + \text{HO}_2$
 18. $\text{ROO} + \text{CH}_3\text{O}_2 = \text{RO} + \text{CH}_3\text{O} + \text{O}_2$
 19. $\text{ROO} + \text{ROO} = \text{RO} + \text{RO} + \text{O}_2$
 20. $\text{ROOH} = \text{RO} + \text{OH}$
 21. RO decomposition
 22. Formation of cyclic oxygenates via cyclization
 23. $\text{QOOH} = \text{unsaturated ether} + \text{HO}_2$ (radical site beta to OOH group)
 24. $\text{QOOH} = \text{unsaturated ether} + \text{carbonyl} + \text{OH}$ (radical site gamma to OOH group)
 25. Addition of O_2 to QOOH ($\text{QOOH} + \text{O}_2 = \text{OOQOOH}$)
 26. Isomerization of OOQOOH and formation of peroxides (e.g., hydroperoxybutyl butyrate) and OH
 27. Decomposition of hydroperoxides to form oxygenated radical species and OH
 28. Cyclic oxygenates reactions with OH and HO_2
 29. Decomposition of large carbonyl species and carbonyl radicals

Reaction classes and rate rules have been discussed extensively in past studies (e.g. [29,35,36]). The rate rules used for DBE in this study are taken from the recent study for *n*-butanol [29], in which rate rules for various reaction classes are modified using recently published rates. This approach was validated using quantum chemistry calculations to verify the most stable DBE conformation and to obtain C–H, C–C and C–O bond dissociation energies (BDEs). All geometry optimizations and frequency calculations were performed using the Gaussian09 suite of programs [39]. The CBS-QB3 and G4 composite methods were selected for their reported accuracies of 1.0 and 1.1 kcal/mol [40,41], respectively. Conformational analysis of DBE and bond-scission products was conducted to determine the most stable conformer of each species.

It is shown in Fig. 2 that the calculated C–H and C–C BDEs are highly sensitive to their location relative to the oxygen site. The presence of the central O atom causes the weaker BDEs of the alpha C–H bond and the stronger BDEs of the beta C–H bond. Similar behavior was also observed from CBS-QB3 calculations for *n*-butanol and the calculated BDEs at alpha, beta, gamma and delta sites are consistent with the computed values for *n*-butanol [29]. Therefore, the BDEs calculated at CBS-QB3 method are used to promote the modeling approach in this study.

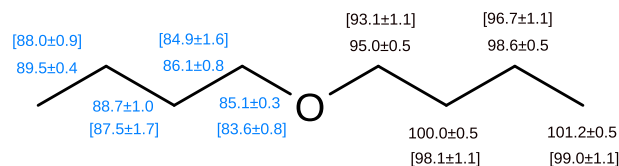


Fig. 2. Schematic diagram of the bond dissociation energies for DBE. The numbers in the brackets are G4 values, and those without the brackets are the CBS-QB3 values. Black values at right side are C–H BDEs, and blue values at left side are C–C or C–O BDEs (kcal/mol at 298 K).

For the sake of brevity, the rate rules proposed in Ref. [29] are not discussed here, so the reader is directed to previous work [29]. A list summarizing the most significant updates to reaction classes and rate rules in the proposed model is presented as follows:

- Hydrogen abstraction rate constants were assigned based on analogous rates used for *n*-butanol [29], except for abstraction by HO_2 radicals that was taken from Ref. [16]. These rates employed by Fischer et al. [16] for DME oxidation increase low temperature reactivity below 600 K.
- H-atom abstractions by OH radicals are amongst the most important in combustion systems. The site-specific rate constants for these reactions were estimated initially from analogous rates proposed for *n*-butanol [29]. The A-factors for each site-specific rate constant were then multiplied by a factor of 1.5 to obtain better agreement with the total rate constant measured by Mellouki et al. [23] for DBE, as shown in Fig. 3. The proposed total rate constant (i.e., solid orange line) obtained by summing the individual rate constants of the alpha, beta, gamma and delta plus OH channels are in good agreement with experimental data below 400 K [24]. Experimental rate constant measurements at higher temperatures are not available for comparison with the proposed rate constants.
- The low temperature mechanism for DBE includes the conventional radical chain branching sequence ($\text{R} + \text{O}_2 \rightleftharpoons \text{RO}_2 \rightleftharpoons \text{QOOH} (+ \text{O}_2) \rightleftharpoons \text{O}_2\text{QOOH} \rightleftharpoons 2\text{OH} + \text{R} + \text{stable}$). DBE is asymmetrical molecule with a central O atom and eight C–H sites that participate in hydrogen migration reactions. The present model allows for hydrogen migration reactions across the central O atom. The rate constants depend on the nature of the C–H bond scission and on the ring strain energy barrier and were assigned based on those proposed in Refs. [29,32]. As DBE has a relatively weak BDE at the alpha C–H site, H-atom abstraction and migration reaction rates are faster at this site. Hence the activation energy for internal H-atom abstraction from the alpha site is decreased by 2 kcal compared to that for a secondary C–H bond

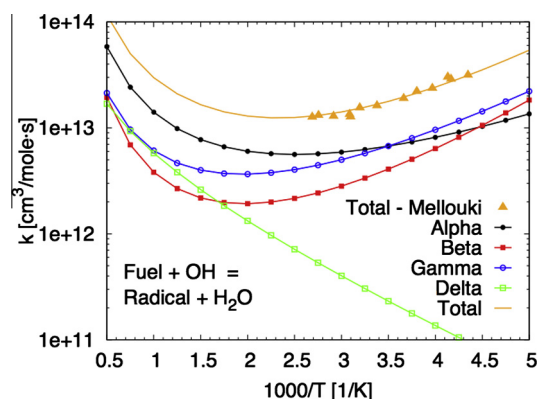


Fig. 3. Rate constants for DBE + OH. Symbols represent total rate constants from Mellouki et al. [24] and lines show rate constants used in this study.

in alkanes [32]. In contrast, the activation energy for internal H-atom abstraction from the beta carbon is increased by 2 kcal. The importance of these reactions for enhancing the low temperature reactivity is discussed later.

- The activation energy for peroxides (e.g., hydroperoxybutyl butyrate) decomposition was reduced to 35 kcal/mol to improve model prediction at low temperatures. This low barrier is consistent with the theoretical calculations of Di Tommaso et al. [22], wherein the hydroperoxymethyl formate derived from diethyl ether was found to decompose rapidly.

3. Experimental facilities

3.1. Laminar flow reactor

A laminar flow reactor (LFR) was used to investigate auto-ignition of DBE in the temperature range of 485–525 K at atmospheric pressure. The concept of the flow reactor is as follows: The fuel is injected in the preheated air flow in the LFR and rapidly mixed. The mixing time is at least 20 times shorter than the ignition delay time [42]. The wall temperature of the LFR matches that of the air/fuel mixture. Therefore, no heat loss of the reactant gas mixture occurs to the surrounding and adiabatic conditions can be assumed for the gas mixture prior to ignition. After a certain time, the mixture auto-ignites leading to a temperature increase. This temperature increase denotes the location of the ignition in the flow reactor, which is used in combination with the information of the preheat temperature and the mass flow rates of all species to calculate the corresponding ignition delay time according to a procedure described in Wada et al. [43]. The temperature increase leads further to a difference in temperature between the gas mixture and the reactor wall introducing heat loss. As the evaluation of the DBE model is performed at low temperatures, the presence of the heat loss arrests any ignition activity after the first stage. Therefore, the ignition delay times obtained with this setup are those of the first stage of DBE/air mixtures.

The LFR is made of stainless steel, has an inner diameter of 20 mm, a length of 5.8 m and is placed in a heated cavity that supplies a homogeneous temperature environment for the flow reactor. Preheated air is introduced into the LFR and DBE is injected through a specially designed fuel injector placed within the main pipe at a distance of 0.15 m from the inlet. The fuel injector has a cylindrical shape with an outer diameter of 10 mm, conical tips at both ends and a total length of 60 mm. The fuel is injected through 12 small holes evenly distributed on the circumference of the fuel injector. It is rapidly mixed with the incoming air flow and the time needed to obtain a homogeneous mixture is short compared with the ignition delay times measured with the setup [42]. The temperature profile over the length of the reactor is monitored using 27 thermocouples inserted approximately 1 mm into the reactor tube. These thermocouples are used to detect the location of the ignition. Using varying volumetric flow rates, the ignition is moved to a location in the reactor where the thermocouple spacing is only 6 cm. This yields an error of 1–2% in the determination of the ignition location at the considered volumetric flow rate ($\nu_{\text{batch}} = 2 \cdots 4 \frac{\text{m}^3}{\text{s}}$). The ignition location is determined using the thermocouple with the highest temperature. The highest temperature gradient occurs between this thermocouple and the one prior to it for all performed measurements. Therefore, the location of the ignition is determined with an error of 1–2% due to the thermocouple spacing. Figure 4 depicts a schematic of the LFR configuration.

Note that the current experimental facility uses the temperature increase of the mixture for the detection of the ignition. Hence, a certain fuel loading is needed. However, the temperature release is relatively small (generally around 20 K), as only first

stage of ignition is monitored. Due to the setup of the experiment, the ignition process is further interrupted after the first temperature increase and only a small part of the fuel is converted which limits the species gradients as well.

The Reynolds (Re) number of the gas mixture is between 612 and 2031 for volume flows between 10 and 35 l/min. With $\text{Re} < 2300$, laminar flow prevails for all LFR measurements. The error of ignition delay measurements is determined to be within 20%. Further detailed descriptions of the experimental apparatus, the measurement technique, the error analysis and the evaluation strategy may be found in [42,43].

3.2. Stagnation flow configuration

Laminar flame speed, S_u^0 , of DBE/air mixtures were measured over a wide range of equivalence ratios, ϕ , in order to provide additional validation targets for the kinetic model. The measurements were carried out in the stagnation flame configuration (e.g. [45]) at atmospheric pressure and for an unburned mixture temperature $T_u = 373 \text{ K}$, for which the fuel partial pressure is below its vapor pressure for all ϕ 's. The diameter of the burner and the separation distance between burners were set at 14 mm. The axial flow velocity was measured using particle image velocimetry (PIV) method with silicon oil droplets as flow tracers.

To determine S_u^0 , axial flow velocities were measured along the stagnation streamline (e.g., [46,47]). The minimum velocity upstream of the flame was defined as the reference flame speed, $S_{u,\text{ref}}$, and the absolute value of the maximum velocity gradient in the hydrodynamic zone was defined as the imposed strain rate, κ , (e.g., [47]). By monitoring the $S_{u,\text{ref}}$ vs. κ variation, S_u^0 could be determined through non-linear extrapolations to $\kappa = 0$ using computationally-assisted extrapolations [48,49]. The error of velocity measurements was determined to be within $\pm 3\%$. The sampling errors in $S_{u,\text{ref}}$ and κ were considered, and the 2σ standard deviations of S_u^0 were determined and are reported in the corresponding figures.

4. Results and discussion

In this section, the proposed kinetic model is validated through comparisons against the experimental results. Sensitivity analysis and reaction pathway analysis were conducted to elucidate the oxidation of DBE at the conditions considered in this investigation. In order to understand better the effects of the ether functional group on ignition of hydrocarbon species, kinetic modeling of ignition delay times and laminar flame speeds for mixtures of oxidizer with fuels structurally similar to DBE were performed and compared to those of DBE.

4.1. Ignition delay times

4.1.1. Experimental determination of ignition delay

LFR experiments were performed for lean and stoichiometric undiluted DBE/air mixtures at atmospheric pressure and temperatures of 485 K to 525 K. As indicated by its high cetane rating, DBE was found to be a highly reactive fuel, with ignition delay times below 1000 ms for temperatures above 500 K, which is more than an order of magnitude faster than for DME [43].

The first stage ignition delay times of the DBE/air mixtures were not measured directly, but obtained using recorded temperature profiles along the LFR and the inlet flow rates as described above. Figure 5 depicts such a temperature profile for a stoichiometric DBE/air mixture at a flow rate of 35 standard liter/min and a preheat temperature of 500 K. The ignition location can be detected using the measured temperature rise, and in Fig. 5 this corresponds

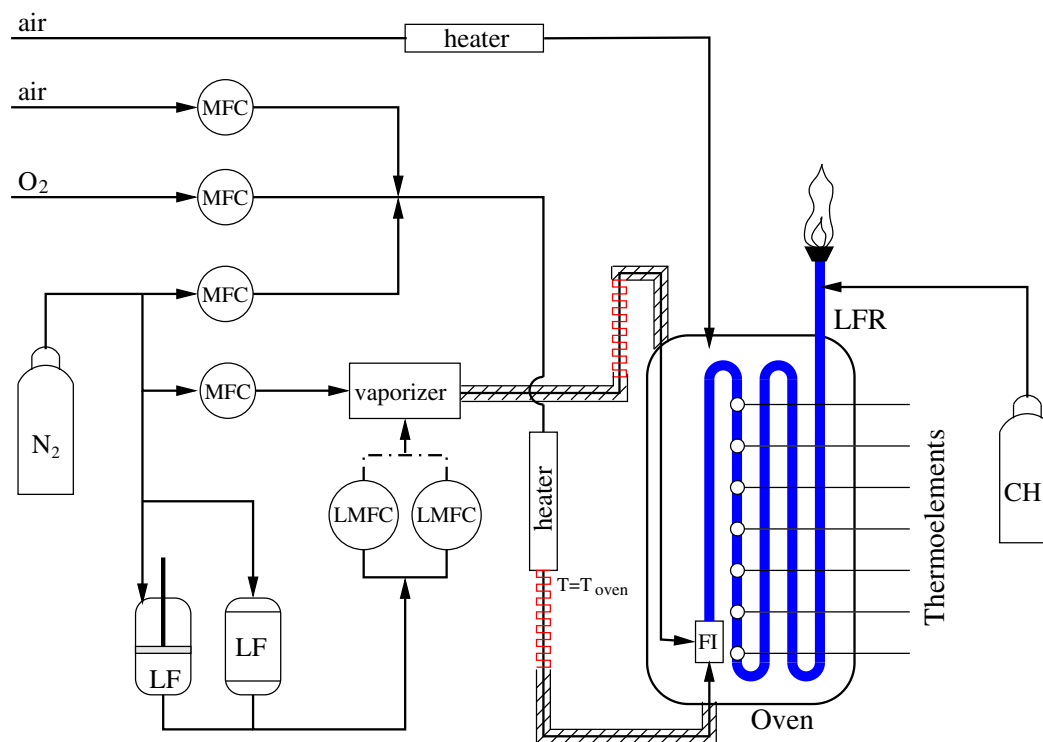


Fig. 4. Schematic of the laminar flow reactor with mass flow controllers (MFC, LMFC), liquid fuel supply (LF), heaters, fuel injector (FI) and thermocouples.

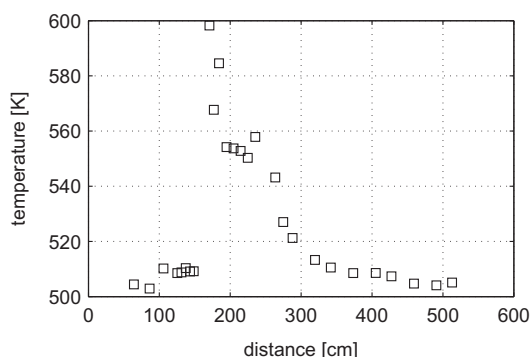


Fig. 5. Temperature profile for a stoichiometric DBE/air mixture at a flow rate of 35 l/min and a preheat temperature of 500 K.

to a distance in the reactor of 170 cm. The decreasing temperature after ignition is due to heat losses of the reactant gas mixture to the oven and resulting freezing of the oxidation reactions. The distance in LFR between mixing and ignition is used to determine the ignition delay time, τ_{ign} , using an average velocity:

$$\tau_{\text{ign}} = \frac{\Delta x_{\text{ign}}}{v_{\text{av}}} \quad \text{with } v_{\text{av}} = 1.5v_{\text{batch}}. \quad (1)$$

Applying analytical solutions for a plug flow and a LFR model, it was shown in [43] that the calculation of the ignition delay times can be performed using this average velocity for laminar flows in LFR. Wada et al. [43] used an analytical approach for the evaluation of two-dimensional effects in the LFR and established a one-dimensional data evaluation approach that accounts for differences between 1D and 2D analysis. However, the calculations are time-consuming due to the diameter/length ratio of the reactor tube (20 mm/5.8 m). Hence, the approach developed by Wada et al. [43] was adopted for the data evaluation.

4.1.2. Kinetic modeling simulations of flow reactor ignition delay times

The numerical calculations of τ_{ign} 's were performed using the zero-dimensional configuration of the constant pressure homogeneous reactor in the FlameMaster code [38]. The computed τ_{ign} is defined as the time to reach a temperature increase of 20 K. The obtained results from the present model are shown in Figs. 6(a)–(d) and compared with the measurements for $\phi = 0.5$, 0.7, and 1.0 at atmospheric pressure. The temperature range for the experimental data is limited by the accessible volume flows of gas mixture in the experimental setup.

The results of Fig. 6(a) indicate that using the present model, τ_{ign} 's of stoichiometric DBE/air mixtures are predicted fairly well over the experimental range. The observed deviations between simulations and experiments are within the reported experimental uncertainty. τ_{ign} 's of lean undiluted DBE/air mixtures are shown in Figs. 6(b) and (c). Good agreement between experimental and computed values can be seen for the lean mixtures as well.

The results of Fig. 6(d) show that the effect of ϕ on the first stage τ_{ign} is minor. This is captured well by the kinetic model, which yields almost identical results for different ϕ 's. Such a behavior was also shown by Mittal et al. [15] for the ignition of DME/O₂/N₂ mixtures in a rapid compression machine and in the detailed chemical kinetic studies by Sarathy et al. for butanol isomers [29] and 2-methylalkanes [32]. The influence of ϕ and pressure on first stage ignition is typically small at such low temperatures as demonstrated by Peters et al. [44] for the example of n-heptane and by Wada et al. [43] for DME. Therefore, their influence on the overall τ_{ign} also vanishes in the low temperature regime, in which the overall ignition delay time approaches the first stage τ_{ign} . Overall, the kinetic model presented here is capable of predicting the experimental τ_{ign} 's accurately.

4.1.3. Low temperature reaction path and sensitivity analysis

Figure 7 illustrates the reaction path analysis for stoichiometric DBE/air combustion at atmospheric pressure and a temperature of 520 K. The reaction fluxes are given at 30% fuel consumption. The

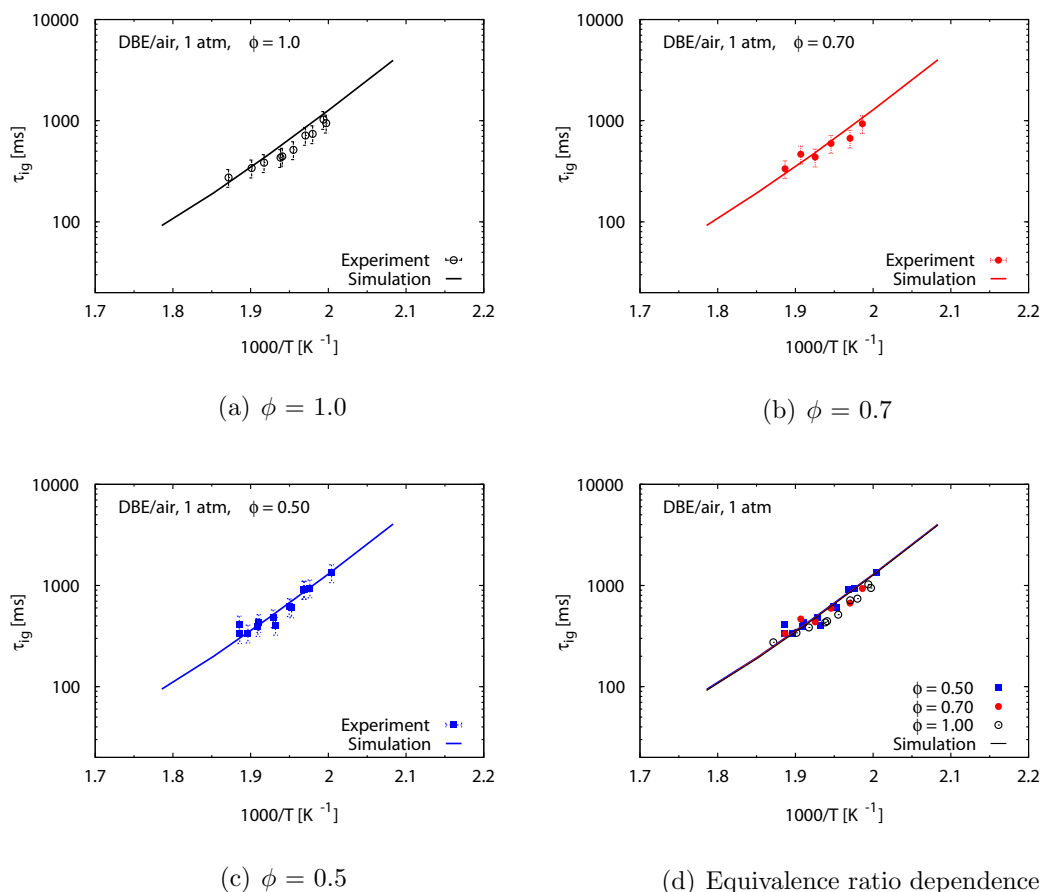


Fig. 6. Ignition delay times of DBE/air mixtures at 1 atm.

oxidation of DBE is initialized via hydrogen atom abstraction from the fuel by molecular oxygen. Due to its slight contribution on the overall consumption of DBE, this pathway is not shown in the pathway diagram. As demonstrated in Fig. 7, DBE is mainly consumed by the build-up of OH. Through the abstraction of an H atom, four primary radicals are produced. It can be observed that the subsequent combustion chemistry of the four primary radicals is significantly different. While the alpha radical ($C_4H_9OC_4H_8-a$) can either decompose directly to 1-butyl radical (C_4H_9) and butanal (C_3H_7CHO) due to the weak C–O bond, or it can be oxidized by oxygen addition to yield a peroxy radical, the beta ($C_4H_9OC_4H_8-b$) and the gamma radicals ($C_4H_9OC_4H_8-c$) combine with oxygen to form peroxy radicals. The decomposition of alpha radicals leads to a chain terminating pathway, and thus inhibits the oxidation reactivity in the low temperature regime. It is seen clearly in Fig. 7 that the chain-branching channel of the peroxy radicals is dominant for the DBE oxidation with a low initial temperature. This low temperature combustion chemistry is demonstrated here using the pathway of the gamma radical ($C_4H_9OC_4H_8-c$) as an example. The addition of molecular oxygen to this radical $C_4H_9OC_4H_8-c$ leads to the formation of a $C_4H_9OC_4H_8-cO_2$ peroxy-radical. After the following isomerization and the second O_2 addition, the peroxy-butoxybutyl-hydroperoxide radical ($C_4H_9OC_4H_8-cO_2H-aO_2$) is produced, which isomerizes rapidly and then decomposes to give an OH radical and a keto-hydroperoxide ($C_4OC_4KETc-a$). The important intermediate species, ketohydroperoxide, can decompose further to form the second OH radical, which completes low temperature chain branching process. At the considered condition, delta radicals ($C_4H_9OC_4H_8-d$) can combine rapidly with oxygen and form peroxy radicals, which initializes a chain branching pathway

and accelerates the ignition of DBE. However, the isomerization of delta radicals can take place as well by transferring H-atoms from alpha carbon sites to the radical site. It should be mentioned here that the cyclic ether formation via cyclization of a hydroperoxy radical plays also a major role in the pathways of beta and delta radicals. As shown in Fig. 8, beta and delta peroxy radicals mainly isomerize to form hydroperoxy radicals. While alpha and gamma hydroperoxy radicals are oxidized directly (Fig. 7), a large number of beta and delta hydroperoxy radicals can cyclize to form a cyclic ether plus an OH radical, which is a chain propagation pathway that competes with the critical chain branching channel. Cyclic ethers are consumed via H atom abstraction by OH and HO_2 radicals, which are the predominant radicals at low and intermediate temperatures. The subsequent cyclic ether radicals then undergo beta-scission to yield smaller hydrocarbon and carbonyl species. As only one OH radical is produced through the cyclic ether formation, this reaction pathway is chain propagating and slightly retards the low temperature oxidation. The alpha and gamma hydroperoxy radicals tend not to cyclize because both these hydroperoxy radicals are formed via 6-membered isomerizations from the alpha C–H site; their cyclization would involve 4-membered transition state rings, which have a higher energy barriers. However, the beta and delta hydroperoxy radicals are formed via 7-membered isomerizations from the alpha site; cyclic ether formation from these radicals is competitive because 5-membered transition state rings have lower energy barriers.

It is interesting to comment on the contribution of H-atom migration reactions (e.g., $RO_2 \rightleftharpoons QOOH$, etc.) across the central O atom towards the low temperature chain branching sequence. DBE has an alpha C–H site with a relatively weak bond dissociation

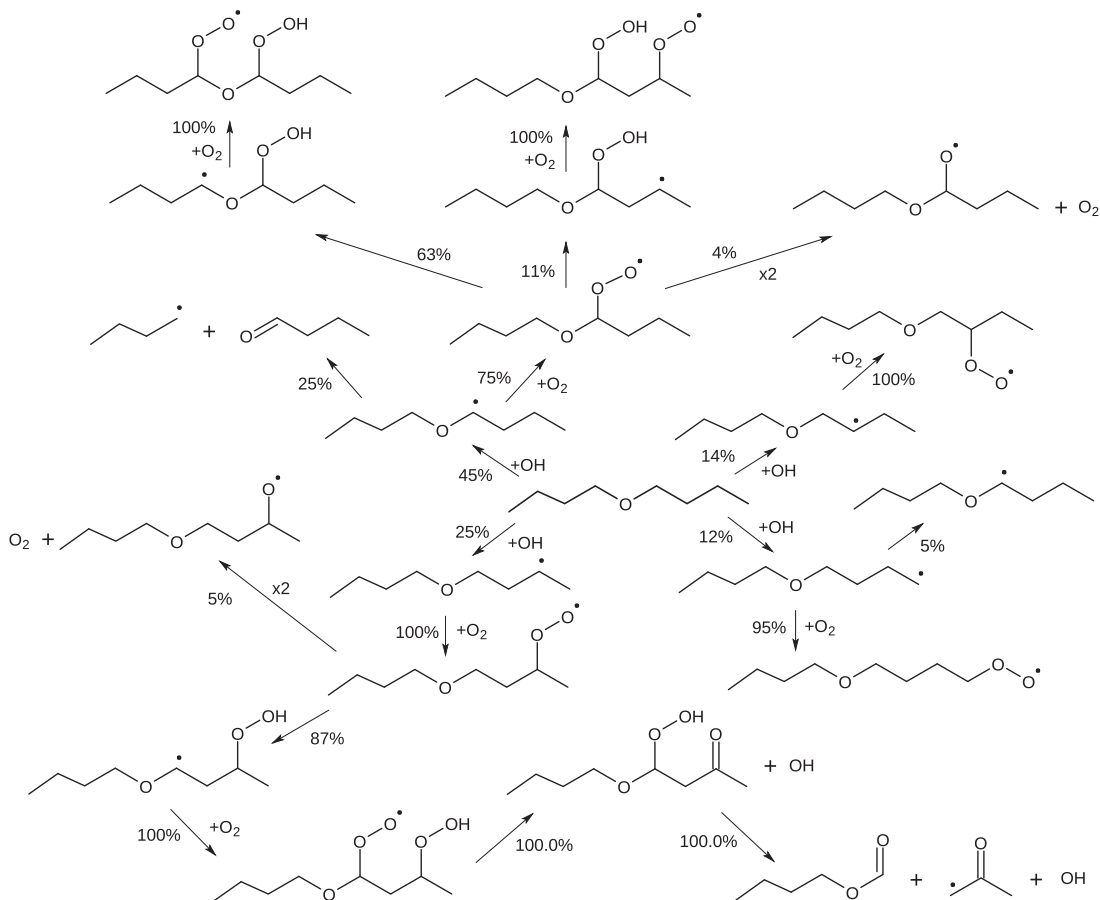


Fig. 7. Reaction path analysis for DBE/air mixture at 30% fuel consumption and at 520 K, 1 atm and $\phi = 1.0$.

energy (BDE), and thus H-atom abstraction and migration reaction rates are faster at this site (e.g., see Fig. 3 DBE + OH rate constants). Fig. 7 indicates that the alpha radical reacts with O_2 to form an RO_2 molecule, which then undergoes a rapid six-membered ring internal H-atom migration reaction with the alpha C–H site on the other side of the central O atom. This eventually results in low temperature radical chain branching leading to ignition. Figure 8 indicates that a similar reaction is important for the beta radical, wherein its RO_2 derivative undergoes a seven-membered ring internal H-atom migration reaction with the opposite alpha C–H site; recall that

these reactions are facilitated due to the weak BDE of the alpha C–H bond. Thus, the present model is able to accurately capture the low temperature ignition characteristics of DBE because it accounts for the weaker BDE of the alpha C–H bond and allows for hydrogen migration reactions across the central O atom. These results are consistent with the theoretical studies of Di Tommaso et al. [22] and Marrouni et al. [25].

To better understand the controlling DBE kinetics, a sensitivity analysis for first stage τ_{ign} 's was performed. The, in essence, logarithmic sensitivity coefficients, LSC, of τ_{ign} are defined as

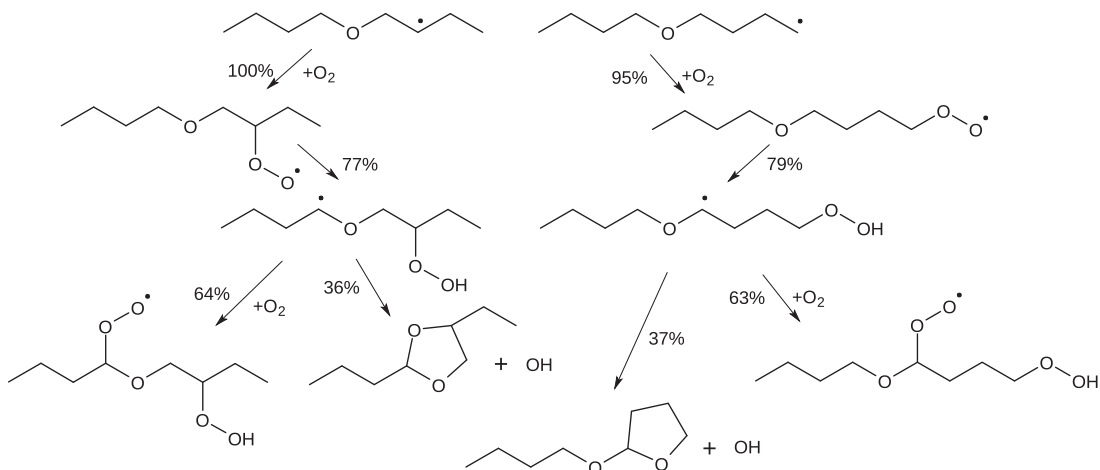


Fig. 8. Reaction path analysis for beta and delta fuel radicals at 30% fuel consumption and at 520 K, 1 atm and $\phi = 1.0$.

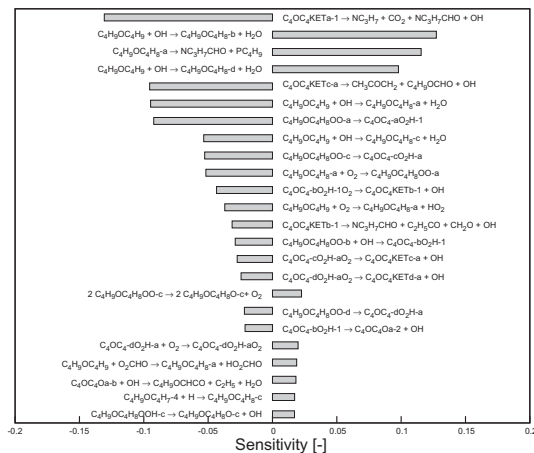


Fig. 9. Logarithmic sensitivity coefficients of ignition delay times for DBE/air mixture at 520 K, 1 atm and $\phi = 1.0$.

$$LSC = \frac{A_{i,0}}{A_i - A_{i,0}} \frac{\tau_{ig,0} - \tau_{ig,i}}{\tau_{ig,0}}, \quad (2)$$

where $\tau_{ig,0}$ is the ignition delay time calculated using the present model and $A_{i,0}$ is the nominal value of the pre-exponential factor of the rate coefficient in the model. τ_{ig} indicates the ignition delay time calculated using A_i , which is obtained by multiplying $A_{i,0}$ with a specified factor. Here, a factor of two is employed. A negative LSC characterizes reactions which enhance the reactivity of the fuel, while reactions with a positive LSC diminish the reactivity.

Figure 9 shows the results of the sensitivity analysis for stoichiometric DBE/air combustion at the condition for which the reaction path analysis was conducted. It is observed that the results from the sensitivity analysis are consistent with those from the path analysis. As found in the reaction path analysis, the chain branching pathway of fuel radicals are dominant at this condition. The H-atom abstraction from the fuel by OH radicals, the addition of O_2 to alpha radicals, the isomerization of alpha and gamma peroxy radicals, the beta-scission of peroxy-butoxybutyl-hydroperoxide radicals ($C_4H_9OC_4H_8-bO_2H-1O_2$) and the decomposition of keto-hydroperoxides ($C_4OC_4KETa-1$ and $C_4OC_4KETc-a$), typical for the low temperature combustion chemistry, are the most sensitive reactions here. As previously discussed, the formation of beta and delta radicals could lead to a chain propagating process, while the decomposition of alpha radicals initializes a chain termination. These reactions compete with the chain branching pathway and hence decelerate ignition. This is confirmed in the sensitivity analysis, as τ_{ign} exhibits large LSC's for these three reactions.

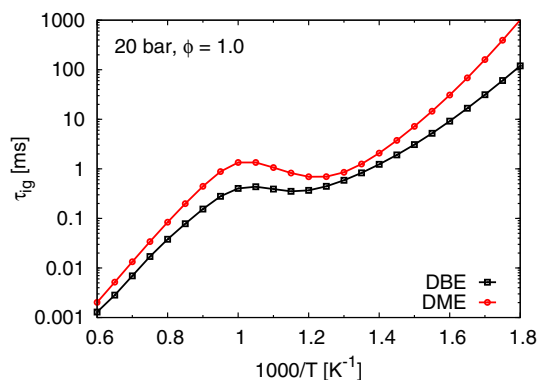


Fig. 10. Ignition delay times of DME/air and DBE/air mixtures at 20 bar and $\phi = 1.0$.

4.1.4. Kinetic modeling simulations of shock tube ignition delay times

In order to gain further insight into the ignition behavior of DBE, numerical simulations with the proposed model were performed for a shock tube configuration and then compared with the results for species having a similar molecular structure, e.g. DME, *n*-octane and *n*-butanol. In addition, this contributes to an improved knowledge about the effects of the long chain ether functional group on auto ignition.

Figure 10 shows the calculated τ_{ign} 's for stoichiometric DBE/air and DME/air mixtures at a pressure of 20 bar. It can be seen readily that the results for DBE exhibit a NTC behavior. For the simulation of DME, the kinetic model proposed by Zhao et al. [13] was employed. As shown in Fig. 10, DBE yields a very similar ignition performance as DME. While the differences of ignition delays between DBE and DME are comparably small in the high temperature range, the differences grow with decreasing temperature. This relatively increased reactivity in the low temperature regime observed for DBE is attributed to the increased chain length that improves the low temperature reactivity of the fuel. One would expect that the ignition reactivity of ethers will not increase further after a critical length of carbon chain is reached, as demonstrated for *n*-alkanes by Westbrook et al. [35] and for 2-methylalkanes by Sarathy et al. [32].

The computed τ_{ign} 's for DBE are compared to those for *n*-octane [32] and *n*-butanol [29] in Fig. 11. Over the entire range of temperatures investigated, DBE exhibits greater ignition propensity compared to *n*-octane, which indicates that the ether functional group in the molecular structure accelerates the oxidation of the fuel. DBE is more reactive than an alkane of the same carbon chain length (e.g., *n*-octane) because of the central O atom. Its presence causes the alpha site C–H bond energy to be lower than that of a secondary C–H bond in an *n*-alkane [22,25]. This results in a facile H-atom abstraction site in DBE, which propels low temperature chain branching via H-atom migrations across the central O atom, as described earlier. Furthermore, the model suggests that peroxides derived from DBE are highly unstable [22] and decompose faster than those derived from alkanes. Compared to DBE, *n*-butanol has an H atom instead of a butyl chain connected directly with the oxygen, which forms an alcohol functional group. The results of Fig. 11 reveal large differences in ignition performance between DBE and *n*-butanol. While DBE exhibits a clear NTC behavior, only a very slight effect of low temperature chemistry on ignition can be obtained starting at around 850 K for *n*-butanol. This difference results from the inclusion of molecular oxygen reactions with a 1-hydroxybutyl radical to form an aldehyde/ketone and HO_2 in the *n*-butanol model [29], which leads to a chain propagating process and thereby inhibits the oxidation reactivity of *n*-butanol. Moreover, the relatively short chain length of *n*-butanol might also be a reason for its low reactivity as the chain branching pathways

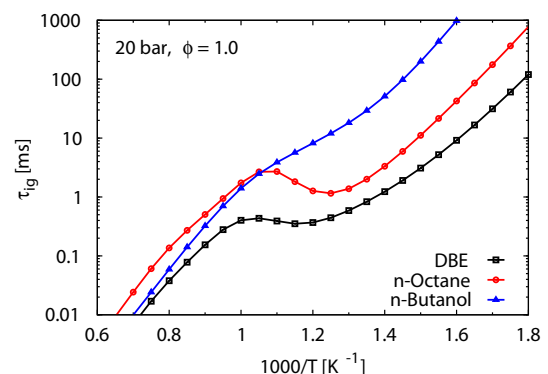


Fig. 11. Ignition delay times of DBE/air, *n*-octane/air and *n*-butanol/air mixtures at 20 bar and $\phi = 1.0$.

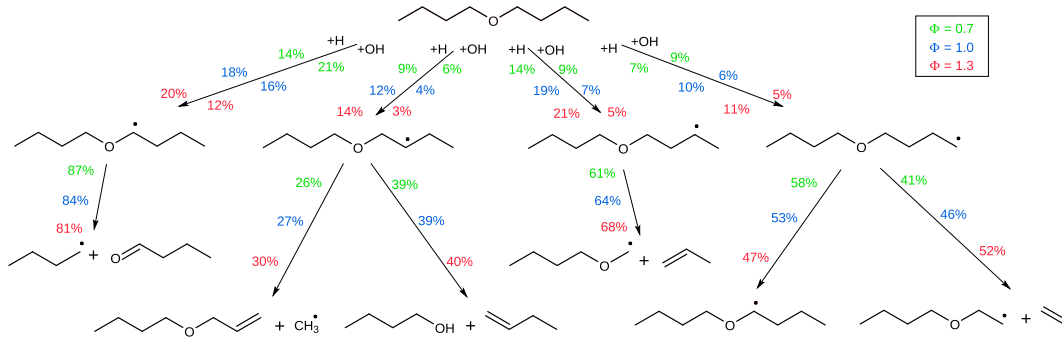


Fig. 15. Reaction path analysis for DBE/air freely propagating flames at 1 atm and $T_u = 373$ K.

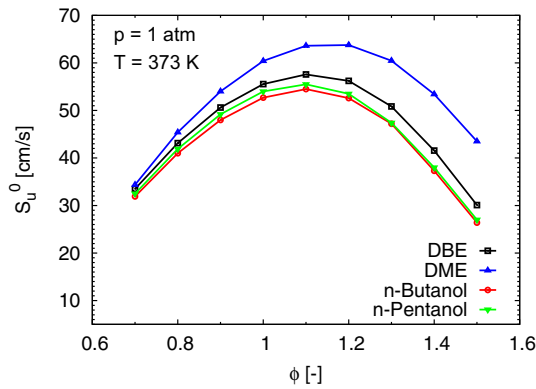


Fig. 16. Computed laminar flame speeds of DME/air, DBE/air, *n*-butanol/air and *n*-pentanol/air mixtures at 1 atm and $T_u = 373$ K.

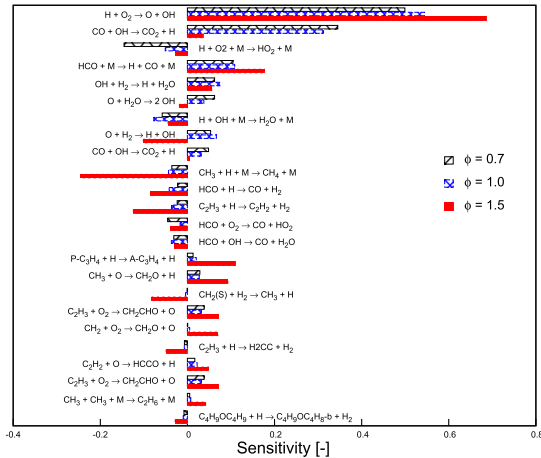


Fig. 17. Logarithmic sensitivity coefficients of laminar flame speed on kinetics for DBE/air mixtures at 1 atm, $T_u = 373$ K and $\phi = 0.7, 1.0$ and 1.5 .

4.2.2. Comparisons of laminar flame speeds of DME/air, *n*-butanol/air and *n*-pentanol/air under similar thermodynamic conditions

S_u^0 's of DME/air, *n*-butanol/air, and *n*-pentanol/air were computed and compared against those of DBE/air mixtures in order to illustrate the effect of molecular structure on flame propagation. The results are shown in Fig. 16, and it can be seen that DME flames propagate notably faster than DBE flames for the majority of ϕ 's considered. On the other hand, DBE flames propagate at rates that are relatively close to those of *n*-butanol and *n*-pentanol flames.

4.2.3. Sensitivity analysis on kinetics

Sensitivity analysis of S_u^0 on kinetics was performed using the FlameMaster software package [38]. LSC's of S_u^0 are defined as

$$\text{LSC} = \frac{A_i}{S_u^0} \frac{\partial S_u^0}{\partial A_i} \quad (3)$$

Here, A_i the pre-exponential factor of each reaction i . The analysis was performed using the present model for $\phi = 0.7, 1.0$, and 1.5 , and the results are shown in Fig. 17 for 1 atm and $T_u = 373$ K.

As expected, S_u^0 exhibits that largest LSC's on the main chain branching reaction between H and O_2 and the main CO oxidation reaction for all ϕ 's. The competing three-body reaction to form HO_2 proves to be one of the main inhibiting reactions. It is notable to identify that S_u^0 is not sensitive to the reaction involving the fuel ($\text{C}_4\text{H}_9\text{OC}_4\text{H}_9$) itself. The most sensitive reaction with fuel is H atom abstraction from the beta site by H radical, which only has a remarkable sensitivity at very rich conditions. As shown in Fig. 17, the prediction of S_u^0 's depends strongly on reactions involving the small hydrocarbon species, e.g. CO, OH, and O radicals similarly to other non-aromatic large molecular weight fuels (e.g. [54–56]).

Sensitivity analysis of S_u^0 on DME kinetics was conducted to investigate the reaction mechanisms which account for the different behavior of DBE and DME. As shown in Fig. 18, the prediction of S_u^0 's depends again strongly on reactions involving the small hydrocarbon species. However, CH_3 channels become more important.

In DME flames, the primary radical (CH_3OCH_2) decomposes to produce formaldehyde and a methyl radical. The formaldehyde then undergoes H-atom abstraction to yield formyl (HCO) radicals that decomposes to provide reactive hydrogen radicals. This reaction pathway involving the principal stable and radical species CH_2O , HCO, and H accelerates flame propagation in rich flames, which is also indicated by Veloo et al. [53] for the example of methanol. However, this reaction pathway is inhibited in DBE flames. While formaldehyde can be directly produced by the fuel radical of DME, the main production channel of CH_2O in DBE flames is the reaction $\text{CH}_3 + \text{O} \rightarrow \text{CH}_2\text{O} + \text{H}$ leading to chain termination.

4.2.4. Sensitivity coefficients on binary diffusion coefficients

LSC's of S_u^0 on binary diffusion coefficients D_{ij} of species i in species j are defined as

$$\text{LSC} = \frac{D_{ij}}{S_u^0} \frac{\partial S_u^0}{\partial D_{ij}} \quad (4)$$

The analysis was computed using the present model for 1 atm, $T_u = 373$ K, and $\phi = 0.7, 1.0$, and 1.5 , and the results are shown in Fig. 19. Similarly to previous studies (e.g. [49,54]), the sensitivity of S_u^0 on fuel-related binary diffusion coefficients is minor. Instead,

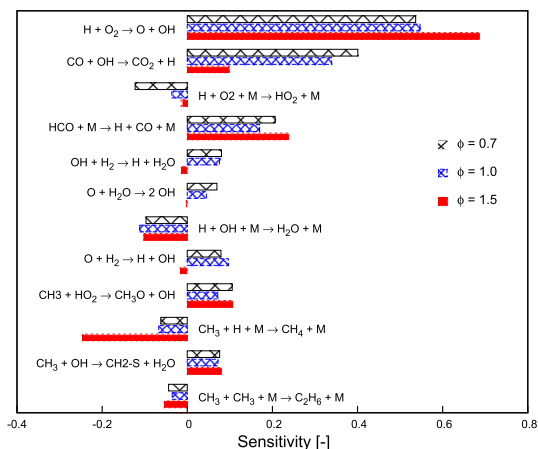


Fig. 18. Logarithmic sensitivity coefficients of laminar flame speed on kinetics for DME/air mixtures at 1 atm, $T_u = 373$ K and $\phi = 0.7, 1.0$ and 1.5 .

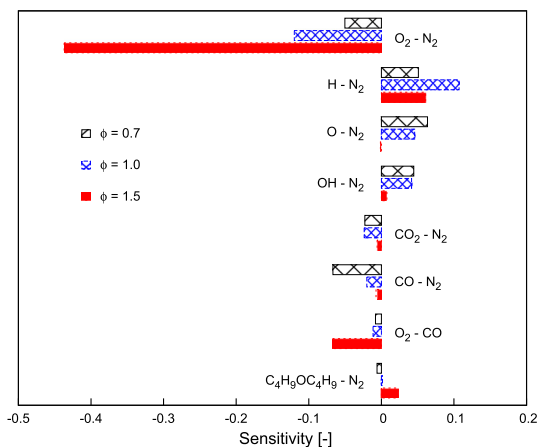


Fig. 19. Logarithmic sensitivity coefficients of laminar flame speed on binary diffusion coefficients for DBE/air mixtures at 1 atm, $T_u = 373$ K, and $\phi = 0.7, 1.0$, and 1.5 .

large LSC's can be seen for species related to the main branching and the main CO oxidation reactions for all ϕ 's.

5. Concluding remarks

A chemical kinetic study was carried out for the first time on the oxidation of di-*n*-butyl-ether. Ignition delay times at low temperatures and laminar flame speeds were experimentally determined in a flow reactor and in the stagnation flame configuration, respectively. A detailed reaction model was developed to describe the DBE oxidation at relevant conditions and the agreement between experimental and numerical results is satisfactory.

The proposed model provides the first global insight into the effects of the long chain ether functional group on ignition delay times. Simulations were performed using the model and the results were compared to those computed for dimethyl-ether, *n*-butanol, *n*-pentanol, and *n*-octane under similar conditions. As indicated also by the cetane number, long chain linear ethers exhibit a higher ignition propensity than normal alkanes and alcohols. The model predicts that the low temperature ignition characteristics of DBE are enhanced because the weaker BDE of alpha C–H bonds, hydrogen migration reactions across the central O atom, and the low barrier of peroxide decomposition all facilitate rapid low temperature radical chain branching reactions.

Additional fundamental combustion experiments of DBE are not available in the literature for further validation of the proposed model. One notable concern is the performance of the model for ignition delay times at intermediate to high temperatures for which validation targets are unavailable. The results of the current investigation provide confidence that the present model is capable of predicting such properties with moderate accuracy. The systematic model development process used in this study has been applied successfully in the past to construct kinetic models of a large number of hydrocarbon and oxygenated fuels and quantum chemistry calculations are also employed to validate the application of the modeling approach. Nevertheless, the model has not been validated against a comprehensive set of experimental data and further investigation should be performed to confirm the model accuracy.

In this sense, future fundamental studies (e.g., experimental ignition data behind shock waves or master equation modeling) will be essential towards improving the model's predictive capability and provide more detailed information of the ether functional group. Experimental studies in jet-stirred reactor will make it easy to compare fuels with different CN values, and also provides species-specific validation data as well.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.10.003>.

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